

# Topic 9.2

## 9.2: An Example of a Template

Here is a program that illustrates how to use a template for dynamic arrays:

```
// MODULE:      dynarr4d.h
// PROGRAMMER:  Gary J. Blair
// LANGUAGE:    C++
// DATE:       96/04/11
//
// PURPOSE:
//   Template for dynamic arrays.
//   Illustrate templates.
//   Note that this header file contains the
//   bodies of the member. This is because the
//   compiler is actually generating code
//   for us.

#include      <iostream.h>
#include      <stdlib.h>

template <class T> class DynamicArray
{
    public:
    DynamicArray( size_t size );
    ~DynamicArray(void);
    T& operator[]( long subscript );
    size_t size(void);

    size_t array_size; // Size of the
                      // dynamic array.
    T *data;           // Pointer to dynamic
                      // memory for the array.
};

// Constructor.

template <class T>
DynamicArray<T>::DynamicArray( size_t size)
{
    long i;    // Loop variable.

    data = new T[size];
    if (! data)
    {
        cerr << "*** Memory allocation failed.\n";
        exit(1);
    }
    array_size = size;
    for (i=0; i<array_size; ++i)
        MODIFICATION: eliminate data[i] = 0;
}

// Destructor.

template < class T > DynamicArray<T>::~~DynamicArray(void)
{
    delete[] data;
    array_size = 0;
}

// Accessor function which returns a
```

```

// reference
// to an element the dynamically-allocated
// memory.

template < class T>
T& DynamicArray<T>::operator[]( long subscript )
{
    if (subscript < 0 || subscript >
        (array_size - 1))
        exit(1);
    return data[subscript];
}

// Inquire the size of the array.

template<class T> size_t
DynamicArray<T>::size(void)
{
    return array_size;
}

// MODULE:      dynarr4d.cpp
// PROGRAMMER:   Gary J. Blair
// LANGUAGE:     C++
// DATE:        96/04/11
//
// PURPOSE:
//   Driver for dynarr4 exercise.
//   Illustrate templates.

#include      <iostream.h>
#include      <stdlib.h>
#include      "dynarr4.h"

void main(void)
{
    DynamicArray<int>      x(12);
    DynamicArray<double>   *q;
    long                  i;

    for (i=0; i<x.size(); ++i)
        x[i] = i * i;

    q = new DynamicArray<double>(12);
    for (i=0; i<q->size(); ++i)
        (*q)[i] = i*i + 0.1;

    for (i=0; i<12; ++i)
        cout << x[i] << "    " << (*q)[i]
              << endl;
}

```

## Output from dynarr4d:

```

0      0.1
1      1.1
4      4.1
9      9.1
16     16.1
25     25.1
36     36.1
49     49.1
64     64.1
81     81.1

```

|     |       |
|-----|-------|
| 100 | 100.1 |
| 121 | 121.1 |

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